

# US Patent & Trademark Office

## Patent Public Search | Text View

United States Patent Application Publication

20250286093

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

SATO; Hiromichi

### FUEL CELL COOLING SYSTEM

#### Abstract

In the fuel cell cooling system, a first cooling flow path, a fuel cell, a second cooling flow path, a radiator, an intercooler, an intermediate heat exchanger, and a bypass flow path, and during power generation of the fuel cell, when the temperature of the refrigerant in the first cooling flow path is lower than the first reference value, a first operation for circulating the refrigerant in the second cooling flow path is performed in a path passing through the intermediate heat exchanger, the intercooler, and the bypass flow path, and when the temperature of the refrigerant in the first cooling flow path is higher than the first reference value, a second operation for circulating the refrigerant in the second cooling flow path is performed in a path passing through the intermediate heat exchanger, the intercooler, and the radiator.

**Inventors:** SATO; Hiromichi (Hadano-shi, JP)

**Applicant:** TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota-shi, JP)

**Family ID:** 1000008279803

**Assignee:** TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota-shi, JP)

**Appl. No.:** 18/964094

**Filed:** November 29, 2024

#### Foreign Application Priority Data

JP 2024-032831

Mar. 05, 2024

#### Publication Classification

**Int. Cl.:** H01M8/04746 (20160101); H01M8/04007 (20160101); H01M8/04186 (20160101);  
H01M8/0432 (20160101); H01M8/04992 (20160101)

**U.S. Cl.:**

## Background/Summary

### CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-032831 filed on Mar. 5, 2024, incorporated herein by reference in its entirety.

### [0002] BACKGROUND

#### 1. Technical Field

[0003] The technology disclosed herein relates to a fuel cell cooling system.

#### 2. Description of Related Art

[0004] A fuel cell system disclosed in Japanese Unexamined Patent Application Publication No. 2002-33108 (JP 2002-33108 A) has a first cooling flow path and a second cooling flow path for circulating a refrigerant. A fuel cell is provided in the first cooling flow path. A radiator is provided in the second cooling flow path. An intermediate heat exchanger is provided across the first cooling flow path and the second cooling flow path. The intermediate heat exchanger exchanges heat between the first cooling flow path and the second cooling flow path.

### SUMMARY

[0005] In the fuel cell system, an intercooler is provided to cool compressed air to be supplied to the fuel cell. In the technology of JP 2002-33108 A, when the intercooler is provided in the second cooling flow path, the refrigerant cooled by the radiator can be supplied to the intercooler and the intermediate heat exchanger. During power generation in the fuel cell, the refrigerant can be circulated in the first cooling flow path, and the refrigerant can be circulated in the second cooling flow path to pass through the radiator, the intermediate heat exchanger, and the intercooler. Heat generated in the fuel cell is transferred to the intermediate heat exchanger by the refrigerant flowing in the first cooling flow path. The intermediate heat exchanger cools the refrigerant in the first cooling flow path by heat exchange with the refrigerant in the second cooling flow path (that is, the refrigerant cooled by the radiator). Therefore, the fuel cell can be efficiently cooled by the refrigerant in the first cooling flow path. The intercooler cools the compressed air by heat exchange with the refrigerant in the second cooling flow path (that is, the refrigerant cooled by the radiator). Therefore, the compressed air can be cooled efficiently.

[0006] When the fuel cell is started in a low-temperature environment, the temperature of the fuel cell is low immediately after the start, and the power generation efficiency is low. When the fuel cell is started, it is necessary to supply the refrigerant to the intercooler. In the fuel cell system described above, when the refrigerant is supplied to the intercooler, the refrigerant is also supplied to the intermediate heat exchanger. Then, the fuel cell is cooled by the refrigerant in the intermediate heat exchanger and the first cooling flow path, and the temperature of the fuel cell is unlikely to increase. Therefore, it is difficult to increase the power generation efficiency of the fuel cell. As described above, when the intercooler is provided in the second cooling flow path, there is a problem in that the temperature of the fuel cell is unlikely to increase in a low-temperature environment. The present specification proposes a technology capable of increasing the temperature of a fuel cell while supplying a refrigerant to an intercooler.

#### First Aspect

[0007] A fuel cell cooling system disclosed herein includes: [0008] a first cooling flow path through which a refrigerant circulates; [0009] a fuel cell provided in the first cooling flow path; [0010] a second cooling flow path through which a refrigerant circulates; [0011] a radiator provided in the second cooling flow path; [0012] an intercooler provided in the second cooling

flow path; and [0013] an intermediate heat exchanger configured to perform heat exchange between the first cooling flow path and the second cooling flow path.

The second cooling flow path includes a bypass flow path provided in parallel to the radiator.

When a temperature of the refrigerant in the first cooling flow path is lower than a first reference value during power generation in the fuel cell, a first operation is performed to circulate the refrigerant in the first cooling flow path along a route in which the refrigerant passes through the fuel cell and the intermediate heat exchanger and circulate the refrigerant in the second cooling flow path along a route in which the refrigerant passes through the intermediate heat exchanger, the intercooler, and the bypass flow path.

When the temperature of the refrigerant in the first cooling flow path is higher than the first reference value during the power generation in the fuel cell, a second operation is performed to circulate the refrigerant in the first cooling flow path along the route in which the refrigerant passes through the fuel cell and the intermediate heat exchanger and circulate the refrigerant in the second cooling flow path along a route in which the refrigerant passes through the intermediate heat exchanger, the intercooler, and the radiator.

[0014] In the fuel cell cooling system described above, when the temperature of the refrigerant in the first cooling flow path is lower than the first reference value during the power generation in the fuel cell, the first operation is performed. In the first operation, the refrigerant circulates through the second cooling flow path along the route in which the refrigerant passes through the intermediate heat exchanger, the intercooler, and the bypass flow path. Since the refrigerant circulates along the route bypassing the radiator, the temperature of the refrigerant in the second cooling flow path is unlikely to decrease. Therefore, the amount of heat transfer between the first cooling flow path and the second cooling flow path in the intermediate heat exchanger is reduced. As a result, the temperature of the refrigerant in the first cooling flow path is unlikely to decrease, and the temperature of the fuel cell can be increased efficiently.

---

## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0015] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0016] FIG. 1 is a block diagram of a fuel cell cooling system;

[0017] FIG. 2 is an explanatory view of the first circulation flow path;

[0018] FIG. 3 is an explanatory view of a second circulation flow path; and

[0019] FIG. 4 is a flowchart of an operation selection process of the fuel cell cooling system.

### DETAILED DESCRIPTION OF EMBODIMENTS

[0020] Following the above aspect 1, additional configurations of the fuel cell cooling system disclosed herein are described below.

#### Embodiment 2

[0021] In the fuel cell cooling system, the first operation, and the second operation according to the first aspect, the refrigerant flows in parallel to the intermediate heat exchanger and the intercooler in the second cooling flow path.

#### Embodiment 3

[0022] In the first operation of the fuel cell and the cooling system according to the first or second aspect, when the temperature of the refrigerant in the first cooling flow path is lower than the second reference value, the circulation speed of the refrigerant in the first cooling flow path is made slower than when the temperature of the refrigerant in the first cooling flow path is higher than the second reference value.

#### Embodiment 4

[0023] In the fuel cell cooling system according to any one of aspects 1 to 3, in the first operation, when the temperature of the refrigerant in the first cooling flow path is lower than the second reference value, the circulation speed of the refrigerant in the second cooling flow path is slower than when the temperature of the refrigerant in the first cooling flow path is higher than the second reference value.

[0024] According to the second aspect, the refrigerant flowing through the second cooling flow path flows into the intermediate heat exchanger and flows into the intercooler. That is, the intermediate heat exchanger exchanges heat between the first cooling flow path and the second cooling flow path, and cools the intercooler.

[0025] The fuel cell cooling system **100** shown in FIG. **1** is mounted on a device (for example, a fuel cell electric vehicle) using a fuel cell as a power source. The fuel cell cooling system **100** includes a first cooling system **10**, a second cooling system **20**, and an intermediate heat exchanger **30**.

[0026] The refrigerant circulates in the first cooling system **10**. The refrigerant circulates in the second cooling system **20**. The intermediate heat exchanger **30** has a first heat exchange flow path **30a** and a second heat exchange flow path **30b**. The first heat exchange flow path **30a** is a part of the first cooling system **10**, and the second heat exchange flow path **30b** is a part of the second cooling system **20**. The intermediate heat exchanger **30** causes heat exchange between the first heat exchange flow path **30a** and the second heat exchange flow path **30b**.

[0027] The first cooling system **10** includes a fuel cell **14** and a first cooling flow path **12**. The fuel cell **14** is provided in the first cooling flow path **12**. The fuel cell **14** is supplied with compressed air from an air compressor (not shown) and hydrogen from a tank (not shown). The fuel cell **14** generates electric power by reacting oxygen with hydrogen, and supplies electric power to a motor (not shown).

[0028] The first cooling flow path **12** has a flow path **12a** and a flow path **12b**. The flow path **12a** is connected to the downstream end of the first heat exchange flow path **30a** and the upstream end of the coolant flow path in the fuel cell **14**. The flow path **12b** is connected to a downstream end of the coolant flow path in the fuel cell **14** and an upstream end of the first heat exchange flow path **30a**. The first cooling flow path **12** includes a flow path **12a**, a refrigerant flow path provided inside the fuel cell **14**, a flow path **12b**, and a first heat exchange flow path **30a**. That is, the first cooling flow path **12** is an annular flow path in which the refrigerant circulates.

[0029] The first cooling system **10** includes a pump **16** and a temperature sensor **18a**, **18b**. The pumps **16** are provided in the flow path **12b**. The pump **16** delivers the refrigerant from the installation position toward the intermediate heat exchanger **30**. When the pump **16** is operated, the coolant circulates in the first cooling flow path **12** in the order of the intermediate heat exchanger **30**, the flow path **12a**, the fuel cell **14**, and the flow path **12b**.

[0030] The temperature sensor **18a** is provided in the flow path **12a**. The temperature sensor **18a** detects the temperature of the coolant flowing into the fuel cell **14**. The temperature sensor **18b** is provided in the flow path **12b**. The temperature sensor **18b** detects the temperature of the coolant discharged from the fuel cell **14** (hereinafter referred to as FC outlet temperature).

[0031] The second cooling system **20** includes a radiator **24**, a second cooling flow path **22**, an intercooler **32**, and a three-way valve **38**.

[0032] The radiator **24** is provided in the second cooling flow path **22**. The radiator **24** cools the refrigerant flowing through the second cooling flow path **22** by heat exchange with the outside air.

[0033] The second cooling flow path **22** includes a flow path **22a**, a flow path **22b**, a flow path **22c**, an intercooler flow path **34**, and a bypass flow path **36**. The flow path **22a** is connected to the upstream end of the second heat exchange flow path **30b** and the downstream end of the coolant flow path in the radiator **24**. The flow path **22b** is connected to the three-way valve **38** and the downstream end of the second heat exchange flow path **30b**. The flow path **22c** is connected to the

three-way valve **38** and the upstream end of the coolant flow path in the radiator **24**. An upstream end of the intercooler flow path **34** is connected to the flow path **22a**. The downstream end of the intercooler flow path **34** is connected to the flow path **22b**. The intercooler flow path **34** is a flow path in which the refrigerant flows in parallel with the intermediate heat exchanger **30**. An upstream end of the bypass flow path **36** is connected to the three-way valve **38**. The downstream end of the bypass flow path **36** is connected to the flow path **22a** upstream of the intercooler flow path **34**. The second cooling flow path **22** includes a flow path **22b**, a flow path **22c**, a refrigerant flow path in the radiator **24**, a flow path **22a**, an intercooler flow path **34**, a bypass flow path **36**, and a second heat exchange flow path **30b**. The second cooling flow path **22** is an annular flow path in which the refrigerant circulates.

[0034] The intercooler **32** is provided in the intercooler flow path **34**. The intercooler **32** cools the compressed air supplied to the fuel cell **14** by heat exchange with the refrigerant flowing through the intercooler flow path **34**.

[0035] The three-way valve **38** switches between a state in which the flow path **22b** is connected to the flow path **22c** and a state in which the flow path **22b** is connected to the bypass flow path **36**.

[0036] The second cooling system **20** includes a pump **26**. The pumps **26** are provided in the flow path **22a**. Further, the pump **26** is provided on the downstream side of the connection portion between the flow path **22a** and the bypass flow path **36**, and on the upstream side of the connection portion between the flow path **22a** and the intercooler flow path **34**. The pump **26** delivers the refrigerant from the installation position toward the intermediate heat exchanger **30** and the intercooler **32**.

[0037] The second cooling system **20** includes a temperature sensor **28a** and a temperature sensor **28b**. The temperature sensor **28a** is provided in the flow path **22a**. The temperature sensor **28a** detects the temperature of the coolant emitted from the radiator **24**. The temperature sensor **28b** is provided in the flow path **22c**. The temperature sensor **28b** detects the temperature of the coolant flowing into the radiator **24**.

[0038] The fuel cell cooling system **100** includes a controller **40**. The control device **40** controls the pump **16**, the pump **26**, and the three-way valve **38** based on the temperature of the refrigerant in the first cooling flow path **12**.

[0039] The fuel cell **14** is supplied with compressed air from an air compressor (not shown) and hydrogen from a tank (not shown). The fuel cell **14** generates electric power by reacting oxygen with hydrogen. During power generation of the fuel cell **14**, the fuel cell cooling system **100** may perform the warm-up control and the normal control described below.

[0040] The warm-up control is executed when the temperature of the fuel cell **14** is low. In the warm-up control, the control device **40** activates the pump **16**. When the pump **16** is operated, the coolant circulates through the first cooling flow path **12** in the order of the intermediate heat exchanger **30**, the flow path **12a**, the fuel cell **14**, and the flow path **12b**. Therefore, the heat generated by the power generation of the fuel cell **14** is transferred to the intermediate heat exchanger **30** (that is, the first heat exchange flow path **30a**) by the coolant in the first cooling flow path **12**. The intermediate heat exchanger **30** transfers the heat generated by the power generation of the fuel cell **14** from the first heat exchange flow path **30a** to the second heat exchange flow path **30b**.

[0041] In the warm-up control, the control device **40** operates the pump **26**. Further, the control device **40** connects the flow path **22b** to the bypass flow path **36** by the three-way valve **38**.

Therefore, the refrigerant circulates in the first circulation flow path **101** indicated by the arrow in FIG. **2**. That is, the refrigerant delivered by the pump **26** flows in parallel to the intermediate heat exchanger **30** and the intercooler **32**. The refrigerant that has passed through the intermediate heat exchanger **30** and the intercooler **32** then flows through the three-way valve **38** to the bypass flow path **36**. Therefore, no refrigerant flows through the radiator **24**. The intermediate heat exchanger **30** cools the refrigerant in the first cooling flow path **12** by heat exchange between the first cooling

flow path **12** and the second cooling flow path **22**. The intercooler **32** cools the compressed air supplied to the fuel cell **14** by heat exchange with the refrigerant flowing through the intercooler flow path **34**.

[0042] As described above, in the warm-up control, the fuel cell **14** is cooled and the compressed air supplied to the fuel cell **14** is cooled by the intercooler **32**. In addition, in the warm-up control, since the refrigerant does not flow through the radiator **24**, the refrigerant in the second cooling flow path **22** is hardly cooled. Therefore, it is difficult to cause heat exchange in the intermediate heat exchanger **30**, and the refrigerant in the first cooling flow path **12** is also difficult to be cooled. Therefore, the temperature of the fuel cell **14** tends to increase.

[0043] Normal control is performed when the temperature of the fuel cell **14** is higher than a reference value. In normal control, the control device **40** activates the pump **16**. Therefore, similarly to the warm-up control, the refrigerant circulates in the first cooling flow path **12**. Therefore, the heat generated by the power generation of the fuel cell **14** is transferred to the intermediate heat exchanger **30** (that is, the first heat exchange flow path **30a**) by the coolant in the first cooling flow path **12**. The intermediate heat exchanger **30** transfers the heat generated by the power generation of the fuel cell **14** from the first heat exchange flow path **30a** to the second heat exchange flow path **30b**.

[0044] In the normal control, the control device **40** activates the pump **26**. Further, the control device **40** connects the flow path **22b** to the flow path **22c** by the three-way valve **38**. Therefore, the refrigerant circulates in the second circulation flow path **102** indicated by the arrow in FIG. **3**. That is, the refrigerant delivered by the pump **26** flows in parallel to the intermediate heat exchanger **30** and the intercooler **32**. The refrigerant that has passed through the intermediate heat exchanger **30** and the intercooler **32** then flows to the radiator **24** via the three-way valve **38**. Therefore, the refrigerant does not flow in the bypass flow path **36**. The radiator **24** cools the refrigerant in the second cooling flow path **22** by heat exchange with the outside air. Therefore, the refrigerant cooled by the radiator **24** flows to the intermediate heat exchanger **30** and the intercooler **32**. The intermediate heat exchanger **30** cools the refrigerant in the first cooling flow path **12** by heat exchange between the first cooling flow path **12** and the second cooling flow path **22**. Since the refrigerant in the second cooling flow path **22** is cooled by the radiator **24**, the intermediate heat exchanger **30** can efficiently cool the refrigerant in the first cooling flow path **12**. The intercooler **32** cools the compressed air supplied to the fuel cell **14** by heat exchange with the refrigerant flowing through the intercooler flow path **34**.

[0045] As described above, in the normal control, the fuel cell **14** is cooled and the compressed air supplied to the fuel cell **14** by the intercooler **32** is cooled. Further, in the normal control, the refrigerant in the second cooling flow path **22** is cooled by the radiator **24**. Therefore, the refrigerant in the first cooling flow path **12** can be effectively cooled by the intermediate heat exchanger **30**. Therefore, the fuel cell **14** can be effectively cooled.

[0046] The control device **40** executes the processing illustrated in the flowchart of FIG. **4**.

[0047] In **S2**, the control device **40** measures the outside air temperature by a temperature sensor (not shown). The control device **40** determines whether the outside air temperature is lower than or equal to the cold determination temperature **T1** (for example, 0° C.). When the vehicle is stopped, the temperature of the refrigerant substantially coincides with the outside air temperature. Therefore, **S2** process is equivalent to determining whether or not the temperature of the refrigerant flowing through the first cooling flow path **12** is equal to or lower than the cold determination temperature **T1**.

[0048] When the outside air temperature is equal to or lower than the cold determination temperature **T1** (YES in **S2**), the control device **40** performs the cold/warm-up control in **S4**. In the cold/warm-up control, the control device **40** performs the above-described warm-up control. In the cold/warm-up control, the control device **40** controls the output current of the fuel cell **14** to a high value. In addition, in the cold/warm-up control, the control device **40** controls the discharge flow

rates of the pumps **16** and **26** to a low value. As described above, in the warm-up control, the temperature of the fuel cell **14** tends to increase. In particular, in the cold/warm-up control, since the output current of the fuel cell **14** is high, heat is likely to be generated in the fuel cell **14**. Further, since the discharge flow rates of the pumps **16** and **26** are low, it is difficult to transfer heat from the fuel cell **14** to the refrigerant. Therefore, the temperature of the fuel cell **14** is particularly likely to increase. As described above, when the air temperature is lower than or equal to the cold determination temperature **T1**, the temperature of the fuel cell **14** is rapidly increased by performing the cold/warm-up control. This increases the power generation efficiency of the fuel cell **14**.

[0049] When FC outlet temperature is lower than the cold/warm-up control termination temperature **T2**, the control device **40** performs cold/warm-up control (that is, NO in YES, **S8** in **S6**) until the power of the vehicles is turned OFF. The control device **40** performs **S12** when FC outlet temperature exceeds the cold/warm-up control termination temperature **T2** (NO in **S6**).

[0050] If the outside air temperature is higher than the cold determination temperature **T1** in **S2**, the control device **40** determines whether FC outlet temperature is lower than the warm-up control termination temperature **T3** in **S10**. That is, the control device **40** determines whether or not the temperature of the coolant flowing through the first cooling flow path **12** is lower than the warm-up control termination temperature **T3**. The warm-up control termination temperature **T3** is a temperature higher than the cold determination temperature **T1**.

[0051] When FC outlet temperature is lower than the warm-up control termination temperature **T3** (YES in **S10**), the control device **40** performs the normal warm-up control in **S12**. In the normal warm-up control, the control device **40** performs the above-described warm-up control. In the normal warm-up control, the control device **40** controls the output current of the fuel cell **14** to a low value (that is, a value lower than **S4**). In the normal warm-up control, the control device **40** controls the discharge flow rates of the pumps **16** and **26** to a high value (that is, a value higher than **S4**). As described above, in the warm-up control, the temperature of the fuel cell **14** tends to increase. However, in the normal warm-up control, since the output current of the fuel cell **14** is low and the discharge flow rates of the pumps **16** and **26** are high, the temperature of the fuel cell **14** is less likely to increase than in the cold-warm-up control. In this way, in a state where the temperature of the fuel cell **14** is high to some extent, the power generation amount of the fuel cell **14** is lowered and the flow rate of the refrigerant is increased more than in the cold/warm-up control, thereby reducing the temperature gradient in the fuel cell **14**.

[0052] When FC outlet temperature is lower than the warm-up control termination temperature **T3**, the control device **40** performs the normal warm-up control until the power of the vehicle is turned OFF (that is, YES, **S14** and NO in **S10**). The control device **40** performs **S16** when FC outlet temperature exceeds the warm-up control termination temperature **T3** (NO in **S10**).

[0053] In **S16**, the control device **40** performs the above-described normal control. Therefore, the fuel cell **14** can be cooled efficiently, and the compressed air can be cooled efficiently by the intercooler **32**.

[0054] When FC outlet temperature is higher than the warm-up control restart temperature **T4**, the control device **40** performs the normal control (that is, NO in NO, **S20** in **S18**) until the power of the vehicle is turned OFF. The warm-up control restart temperature **T4** is a temperature lower than the warm-up control termination temperature **T3**. When FC outlet temperature becomes lower than the warm-up control restart temperature **T4** (YES in **S18**), the control device **40** performs the normal warm-up control in **S12**.

[0055] In the embodiment, the intercooler **32** is provided in the intercooler flow path **34**. However, the intercooler **32** may be provided in the flow path **22b**. The intercooler **32** may be provided between the pump **26** and the intermediate heat exchanger **30**. That is, the refrigerant may flow in series with the intermediate heat exchanger **30** and the intercooler **32**.

[0056] The cold/warm-up control and the normal warm-up control of the embodiment are examples

of the first operation. The normal control of the embodiment is an example of the second operation. [0057] The warm-up control termination temperature T3 of the embodiment is an example of the first reference value. The cold determination temperature T1 of the embodiment is an example of the second reference value.

[0058] In the embodiment, the flow path changing device for changing the flow path of the refrigerant in the second cooling flow path 22 by the three-way valve 38 is configured, the flow path changing device may be configured by other devices. Further, in the embodiment, the three-way valve 38 is a solenoid valve, but the flow path changing device may be configured by a device that does not use electricity. For example, a flow path changing device may be configured by a device that switches a flow path by thermal expansion of a material, such as a thermostat.

[0059] In the embodiment, the temperature sensor 18a detects the temperature of the coolant flowing into the fuel cell 14. However, the temperature-sensor 18a may be provided at or near the outlet or inlet of the intermediate heat exchanger 30.

[0060] The temperature sensor 18b detects the temperature of the coolant discharged from the fuel cell 14. However, the temperature-sensor 18b may be provided at or near the outlet or inlet of the intermediate heat exchanger 30.

[0061] In the embodiment, the temperature sensor 28a detects the temperature of the coolant emitted from the radiator 24. However, the temperature sensor 28a may be provided at or near the outlet or inlet of the intermediate heat exchanger 30.

[0062] In the embodiment, the temperature sensor 28b is provided in the flow path 22c. However, the temperature sensor 28b may be provided at or near the outlet or inlet of the intermediate heat exchanger 30.

[0063] While the embodiments have been described in detail above, these are merely illustrative and do not limit the scope of the claims. The technology described in the claims includes various modifications and alterations of the specific examples described above. The technical elements described in this specification or in the drawings may be used alone or in various combinations, and are not limited to the combinations described in the claims at the time of filing. Further, the technology illustrated in the present specification or the drawings achieves a plurality of objects at the same time, and has technical usefulness by achieving one of the objects.

## Claims

1. A fuel cell cooling system comprising: a first cooling flow path through which a refrigerant circulates; a fuel cell provided in the first cooling flow path; a second cooling flow path through which a refrigerant circulates; a radiator provided in the second cooling flow path; an intercooler provided in the second cooling flow path; and an intermediate heat exchanger configured to perform heat exchange between the first cooling flow path and the second cooling flow path, wherein the second cooling flow path includes a bypass flow path provided in parallel to the radiator, when a temperature of the refrigerant in the first cooling flow path is lower than a first reference value during power generation in the fuel cell, a first operation is performed to circulate the refrigerant in the first cooling flow path along a route in which the refrigerant passes through the fuel cell and the intermediate heat exchanger and circulate the refrigerant in the second cooling flow path along a route in which the refrigerant passes through the intermediate heat exchanger, the intercooler, and the bypass flow path, and when the temperature of the refrigerant in the first cooling flow path is higher than the first reference value during the power generation in the fuel cell, a second operation is performed to circulate the refrigerant in the first cooling flow path along the route in which the refrigerant passes through the fuel cell and the intermediate heat exchanger and circulate the refrigerant in the second cooling flow path along a route in which the refrigerant passes through the intermediate heat exchanger, the intercooler, and the radiator.

2. The fuel cell cooling system according to claim 1, wherein, in the first operation and the second



operation, the refrigerant flows in parallel through the intermediate heat exchanger and the intercooler in the second cooling flow path.

**3.** The fuel cell cooling system according to claim 1, wherein, in the first operation, when the temperature of the refrigerant in the first cooling flow path is lower than a second reference value, a circulation rate of the refrigerant in the first cooling flow path is reduced compared to a case where the temperature of the refrigerant in the first cooling flow path is higher than the second reference value.

**4.** The fuel cell cooling system according to claim 3, wherein, in the first operation, when the temperature of the refrigerant in the first cooling flow path is lower than the second reference value, a circulation rate of the refrigerant in the second cooling flow path is reduced compared to the case where the temperature of the refrigerant in the first cooling flow path is higher than the second reference value.

---